A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 34

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Impact depends on total population x per capita impact **1** With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added **1** Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Urban areas are supported by resources brought in from elsewhere – when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage **1** Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.4	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Climate:** warming increases evaporation, cloud and albedo – higher CO2 increases photosynthesis and sequestration – hydrological cycle: more evaporation gives more precipitation – populations: density-dependent factors return numbers to carrying capacity **up to 2** In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) **up to 2** Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **carrying capacity** → the largest population the area can support **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **abiotic** → non-living / physical factors	1
02.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Deforestation **1** Ploughing of grassland / expansion of farmland **1** Reservoir creation / flooding **1** Mineral extraction, especially open-cast mining **1** Urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins) – wastes of one process are raw materials for another so nothing is depleted or accumulates **1** Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials **1 Human systems are linear:** extract - process - manufacture - use - discard – minerals are dispersed after use or mixed so recovery is difficult **1** Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up – fossil fuel use is inherently linear and unsustainable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life **1** They leach from landfill or are released by incineration – persistent, bioaccumulating **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **incineration** → burning waste, usually with energy recovery • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) **1** Easy identification of materials (coding) and separation of components so they can be recycled **1** Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) **1** Use of recycled and recyclable / biodegradable materials – reduced packaging and material use (Cradle to Cradle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** A larger area supports larger populations, which are less likely to be lost to chance events or inbreeding, and it can hold species with large territories such as top predators **1** A larger area holds a wider range of habitats and microhabitats, so more species can find the conditions they need **1** A compact, rounded shape has the smallest perimeter for its area, so less of the reserve is affected by edge effects such as wind, drier conditions, spray drift and disturbance from outside **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** It allows individuals to move between the two areas, so the populations interbreed as one larger gene pool and inbreeding is reduced **1** An area that loses a species can be recolonised from the other, and species can shift their range as the climate changes rather than being trapped in an isolated fragment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
04.3	**1 mark each** 1 mark for each of the 4 points below. **1 Water levels:** sluices and bunds hold water back to create ponds, scrapes and seasonally flooded marsh, adding wetland habitats and a range of depths for different species **1 Soil nutrient levels:** cutting and removing hay rather than leaving it, and never fertilising, lowers the nutrient level so that vigorous grasses cannot out-compete the wildflowers of species-rich grassland **1 Topography:** banks, scrapes and south-facing slopes create warm and cool, wet and dry microhabitats within a small area, each suiting different species **1 Light levels:** thinning, coppicing and cutting rides and glades lets light reach the woodland floor, so ground flora and the insects that depend on it can grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **biodiversity** → the variety of life: habitat, species and genetic diversity	4

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 3 points below. **1** A plagioclimax is a community held at an earlier stage of succession than its natural climax by continued human activity, so it never develops into woodland **1** Grazing animals eat the tree and shrub seedlings that would otherwise shade out the grassland, so succession is deflected and the open habitat survives **1** The species, stocking density and timing of grazing are chosen to suit the habitat – cattle graze less selectively and their trampling creates bare patches for seedlings, while sheep graze close and produce a short, even sward **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3
04.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Invasive non-native species such as rhododendron or Himalayan balsam out-compete and shade out native species, and have no natural controls here **1** Introduced predators or competitors, such as grey squirrels or mink, reduce native populations directly **1** Without their natural predators, herbivore numbers such as deer can exceed the carrying capacity, preventing tree regeneration and damaging the ground flora **1** Culling may also be needed to keep a population within the reserve's carrying capacity so that it does not damage the habitat the whole community depends on **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-49; AQA spec 3.1.3.3-3.1.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala) – increases evaporation **1** Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase **1 Evapotranspiration changes:** increased in irrigated arid areas, decreased where forests were cleared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **Evapotranspiration** → evaporation plus transpiration **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	3
05.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Seasonal / local:** avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes **2 Organic:** uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species – free-range and Fairtrade also accepted for welfare / social reasons **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** Food shortages after 1945 (dependence on US food aid) – farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock **1 Guaranteed market / price support:** government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output **1 Consequences:** loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution – later food surpluses (butter mountains) that could not be sold without harming LEDC producers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Marks shared** Up to 5 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2** A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds – oak supports many invertebrates **up to 2** B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies – fewer niches **up to 2** C: non-indigenous single-species, single-age conifers support few native invertebrates – dense shade – no clearings or dead wood – needles acidify soil **up to 2** Habitat diversity / structural variety increases the range of niches and hence species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	5
06.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Designated protected areas:** SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks **1 Legal protection of ancient woodland:** planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) **1 Conservation management:** coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer **1** Planting new native woodlands linking fragments (corridors) – Woodland Trust / National Trust ownership **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** No species is completely dominant and food webs have many alternative links **1** So a change in the population of one species has a relatively small impact on the whole community **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Spearman's rank correlation coefficient **1** Mann-Whitney U test **1** Chi-squared test **1** t-test **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Spearman** → a test for correlation between two sets of ranked data • **correlation** → a relationship in which the two variables change together • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	4

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chi-squared compares frequencies / counts in categories **1** Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **chi-squared** → a test comparing observed frequencies with expected frequencies • **t-test** → a test for a difference between the means of two normally distributed sets • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is always some remaining uncertainty / a probability that the result was produced by chance **1** Statistics only give a level of confidence (eg 95%, 99%) that the result is significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Bars drawn above and below each mean showing plus and minus one standard deviation **1** If the bars of two means overlap a lot the difference is probably not significant – if they do not overlap it probably is **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Copying the structural / functional adaptations of living organisms in the design of manufactured items **1** Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use **1** Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) **1** Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) **1** Termite mound convection copied for natural ventilation of buildings – hollow strutted bird bones in lightweight roofs – gecko toe adhesion – burr hooks - Velcro – lotus leaf self-cleaning glass – spider silk for airbags / body armour – bat / dolphin echolocation - ultrasound scanners **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores – in controlled doses these affect human physiology **1 Examples:** morphine / codeine from poppies – quinine from cinchona bark (malaria) – aspirin from willow bark – taxol from yew (cancers) – diosgenin from Mexican yam (contraceptives, cortisone) **1 Animal examples accepted:** AZT from a marine sponge (HIV), shark liver lipids (tumours) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals **1** Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	1 mark for the point below. **1** Squid giant nerve cells - nerve function / heart disease, stroke – marsupials (kangaroo, wallaby pouch) - embryo development – hippo skin secretion - sunscreen / burns – sea urchin embryos - teratogen testing – armadillo - leprosy – marine sponge proteins - transplant rejection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** wild catches have not increased since about 1990 as stocks are fully or over-exploited – aquaculture output growing rapidly and approaching the wild catch – fish are the last large-scale hunted food • **Advantages of aquaculture:** control of limiting factors, selective breeding, gender control, high yields per area – fish have low FCRs (no thermoregulation / support) compared with livestock – herbivores (tilapia, carp) and molluscs very efficient – polyculture, IMTA, aquaponics, rice-fish • **Limits:** carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins) – some species (bluefin tuna) will not breed in captivity – consumer preference for carnivores • **Environmental impacts:** habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control – and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits – MSY management needs good data • **Judgement:** aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing – both need sustainable management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquaculture** → fish or shellfish farming	9
	• **gene pool** → the range of alleles / genes present in a population • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	
10.2	1 mark for the point below. **1** A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Physical features:** mass / gravity retaining an atmosphere and giving pressure for liquid water – distance from the Sun and the Goldilocks zone – speed of rotation reducing temperature extremes – tilt producing seasons – magnetic field deflecting the solar wind – the Moon stabilising the orbit and producing tides • **Importance of liquid water:** solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • **Development of life:** Archaea at vents – photosynthesis – oxygen reacting with iron then accumulating – ozone layer allowing colonisation of land – aerobic organisms • **Life changing conditions:** carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases	25
	• **Methods of studying past conditions:** proxy data, isotope ratios, ice cores, Mg:Ca ratios – understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Ecological features:** cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web – specific abiotic requirements (temperature, light, salinity, low turbidity) • **Importance:** quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • **Vulnerability:** narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching – slow growth so slow recovery – physical damage – overfishing and food-web effects (crown of thorns) – introduced species – dependence on mangroves and seagrass • **Conservation:** MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	

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